

A pre-feasibility study for the Establishment of a Hospital in Ma'an



2017

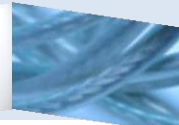


Table of contents

1.	EXECUTIVE SUMMARY	7
	KEY HIGHLIGHT OF JORDAN.....	7
	JORDAN INVESTMENT COMMISSION AIMS TO:	12
2.	MARKET ANALYSIS.....	13
	<i>Project description</i>	13
	<i>Project Objectives</i>	13
	<i>Target segments</i>	14
	<i>Proposed Services</i>	14
	<i>Market Size Analysis</i>	15
	<i>Capacity of the project</i>	17
	<i>Price Analysis and Pricing Policy</i>	18
	<i>Projected Revenues</i>	20
3.	TECHNICAL ANALYSIS	21
	<i>Required Human Resources</i>	21
	<i>Site Analysis</i>	24
	<i>Description of Services</i>	24
	<i>Technical Requirements</i>	25
	<i>1.1.1 Land and Construction</i>	25
	<i>Machinery and Equipment</i>	26
	<i>Furniture</i>	27
	<i>Vehicles and Transportation</i>	27
4.	FINANCIAL ANALYSIS.....	27
	<i>Assumptions</i>	29
	<i>Capital Expenditures</i>	31
	<i>Operating Expenses</i>	35
	<i>General and Administrative Expenses</i>	35
	<i>Marketing Expenses</i>	36
	<i>Human Resources</i>	36

<i>Depreciations.....</i>	<i>42</i>
<i>Loan.....</i>	<i>43</i>
<i>Income Statement.....</i>	<i>44</i>
<i>Expected Cashflows Statement</i>	<i>45</i>
<i>Expected Balance Sheet.....</i>	<i>47</i>
<i>Feasibility Indicators</i>	<i>49</i>
<i>Sensitivity Analysis</i>	<i>50</i>

List of Tables

Table 1 economical indicators	7
Table 2 Population Distribution according to Age Group.....	8
Table 3 Number of treatments by physician in Ma'an Governorate by gender and age group of 2015	15
Table 4 Percentages of Basic Components of Private Hospital Profits.....	16
Table 5 the workload in Ma'an hospitals.....	17
Table 6 capacity of the hospital.....	17
Table 7 sample of surgeries prices.....	18
Table 8 Hospital capacity.....	19
Table 9 projected revenues during the years (2018 - 2027).....	20
Table 10 the required labor for the project	21
Table 11 Required land for the project	25
Table 12 Required Construction Work	25
Table 13 Machinery and Equipment.....	26
Table 14 the required furniture for the project	27
Table 15 Vehicles and Transportation.....	27
Table 16 General Assumptions	29
Table 17 Currency Exchange Rates	29
Table 18 Annual Growth Rates Assumptions.....	29
Table 19 Expenses Assumptions.....	29
Table 20 Income Tax Assumptions	29
Table 21 Weighted Average Cost of Capital.....	30
Table 22 Land Capital Expenditure.....	31
Table 23 Construction Work Capital Expenditure	31
Table 24 Machinery and Equipment Capital Expenditure	31
Table 25 Furniture Capital Expenditure	32
Table 26 Transportation and Vehicles Capex	32
Table 27 Pre Operating Expenses.....	32
Table 28 Working Capital.....	33
Table 29 Expenses Summary.....	33
Table 30 Sources of Funding	33
Table 31 Operating Expenses	35

Table 32 General and Administrative Expenses	35
Table 33 Marketing Expenses	36
Table 34 Expected staff preparation.....	36
Table 35 Expected monthly salaries (2018-2027)	38
Table 36 Capex and Depreciation Expenses	42
Table 37 Depreciations and Additions on Construction Works	43
Table 38 Loan Details	43
Table 39 Income Statement.....	44
Table 40 Expected Cashflows	46
Table 41 Expected Balance Sheet.....	48
Table 42 Free Net Cash flows Table	49
Table 43 payback period	50
Table 44 Financial Analysis Results.....	50
Table 45 Sensitivity Analysis	51

Table of Figures

Figure 1 Population of Jordan.....	8
Figure 2 Sectors Contribution to Jordan's Economy.....	9

1. Executive Summary

The project aims to establish an advanced private hospital in Ma'an (216 km south of Amman). It will be a distinguished medical hospital with 115 beds, containing all specialties and high medical quality.

The idea was presented to contribute to the provision of a high-quality medical environment in the governorate and to reduce overcrowding in other hospitals and health centers. In addition, the implementation of this project will attract medical tourism to those who are coming to the Kingdom for treatment from various countries, especially from the Gulf region and Yemen who are currently heading to the capital because of the lack of high-level medical personnel in the province.

Table 1 economical indicators

Indicator	
Total Investment Cost (Jordan Dinars)	22,967,229
Internal Rate of Return (IRR)	%21.33
Pay-Back Period (Years)	5
Net Present Value (Jordan Dinars)	10,207,500
Job Opportunities provided by the Project	168
Total Revenues (For 10 Years)	159,931,625
Total Operating Expenses (For 10 years)	91,480,889
Total Net Profit (For 10 years)	30,711,726

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

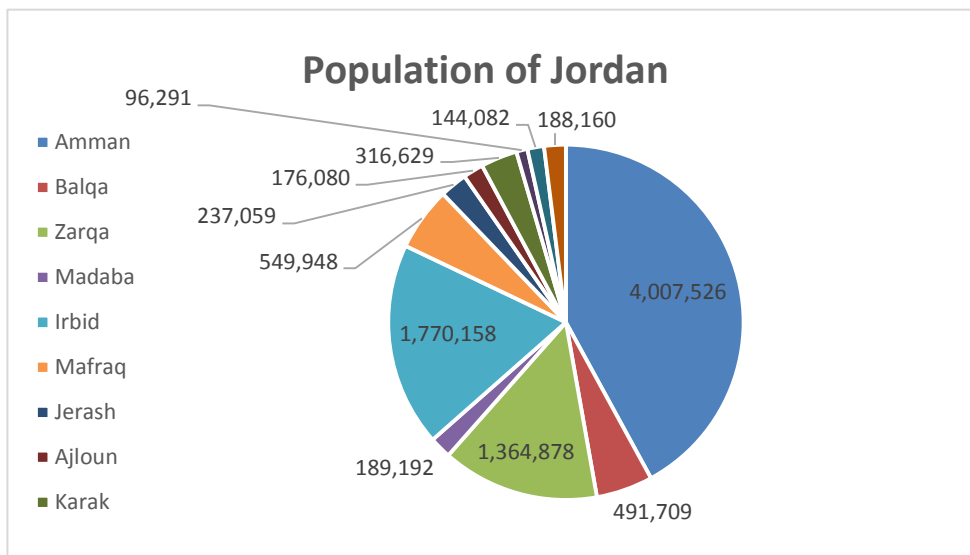
Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the

Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

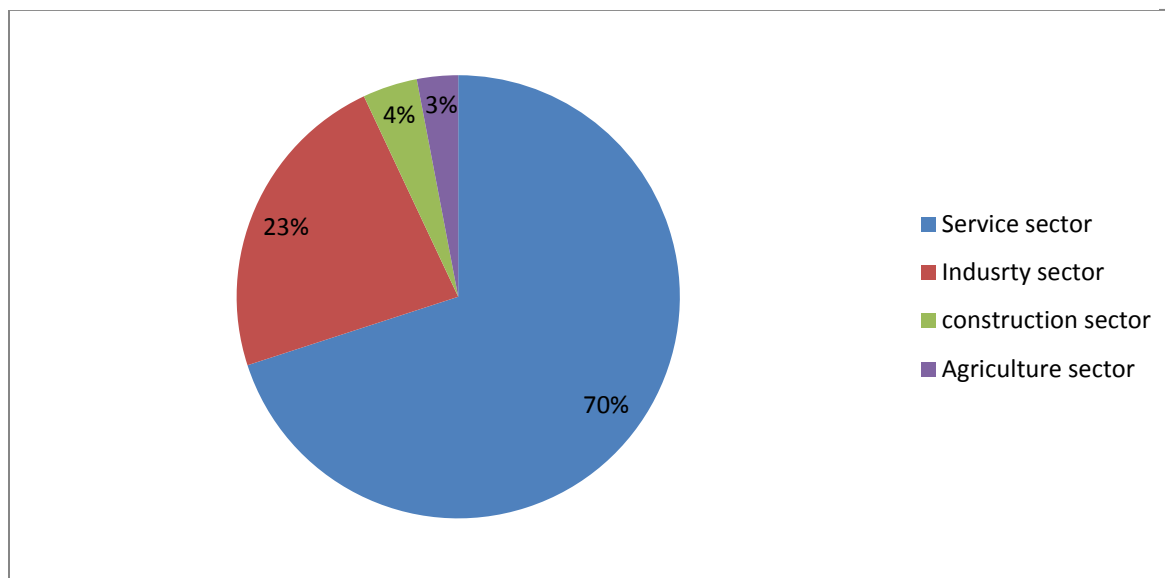
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the

Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally. The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and

legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

The investment environment in Ma'an:

As per in the Investment Law No. 30 of 2014 and the Income Tax Reduction Law in the Less Developed Areas (issued under Article (5) of the Investment Law No. (30) Of 2014), the following taxes and exemptions are imposed on investment projects:

Table 3 payable income tax and tax exemptions

Development areas	Less developed areas
Income tax is 5%	Value of income tax 20% According to the divisions in the law, Ma'an governorate is classified as category B where investors get 80% tax exemption on income tax for 20 years from the date of investment.

2. Market analysis

Project description

The project aims to establish an advanced private hospital in Ma'an (216 km south of Amman). It will be a distinguished medical hospital with 115 beds, containing all specialties and high medical quality.

The idea was presented to contribute to the provision of a high-quality medical environment in the governorate and to reduce overcrowding in other hospitals and health centers. In addition, the implementation of this project will attract medical tourism to those who are coming to the Kingdom for treatment from various countries, especially from the Gulf region and Yemen who are currently heading to the capital because of the lack of high-level medical personnel in the province.

Project Objectives

The objectives of the project are summarized as follows:

- Using the available medical staff in the governorate.
- Meeting the high demand for specialized services in all fields.
- The competitive advantage of the health care sector in the Kingdom compared to the Gulf countries.
- Achieve a profitable return to the entrepreneur.

Target segments

- The local in the governorate and the people of the southern governorates.
- Arab and international communities seeking treatment.

Proposed Services

The proposed project aims to establish a private hospital. So as to serve patients from Jordan and other countries. Basic internal facilities include emergency rooms, ICU, meeting rooms, operating rooms suitable for performing major types of surgeries. It will also include a clinic, a laboratory and a radiology department for a full medical examination.

The hospital consists mainly of the following main departments:

- Emergency department.
- Operating rooms.
- Department of Surgery.
- Cardiology Department.
- Intensive Care Department (ICU-CCU).
- Orthopedic department.
- Internal Medicine Department.
- Department of Obstetrics and Gynecology.
- Children Department.
- Laboratory and blood bank.
- Radiology Department.
- Diagnostic services.
- Sterilization Department.
- Pharmacy.
- Personnel, Finance Section.
- Department of General and Medical Maintenance.
- Department of Medical Records.
- Infection control department.
- IT department.

- Nutrition Department.
- Department of Nursing.
- Medical Administration.

The project will serve the local population in the governorate, the south and the foreigners. The Jordanian government and private hospitals have begun to realize Jordan's unique position in the provision of healthcare at competitive prices, which will make Jordan a center of health care and therapeutic tourism in the region. So that the hospital is committed to applying the approved national and international standards for hospitals. And to provide specialized medical services through the use of modern technology and employ high-quality, as well as respect for the rights of patients and their families, values and beliefs, taking into account the priority of their safety and satisfaction.

The project will also have various incentives provided by the Jordanian government due to the nature of the project, such as income tax exemptions and social services, as well as exemptions on customs duties, which will provide a convenient financial base for the project during the first 10 years of operation.

Market Size Analysis

The proposed location of this project in Ma'an will give patients the opportunity to receive various treatments, such as orthopedic, cardiology, surgery and other treatments. The hospital will also provide a comprehensive range of services for all patients, both outpatient and inpatient clinics. These services will include diagnosis, treatment, preventive measures, rehabilitation, pediatric cardiology and emergency services.

Table 3 Number of treatments by physician in Ma'an Governorate by gender and age group of 2015

categories	Less than a year	1-4	5-14	15-44	+45	Total	Total for both genders
Male	11.55	17717	24536	38416	34411	126132	253274
Female	10570	17338	23910	29562	35759	127139	

Jordanian Ministry of Health

The treatment is not directly linked to any demographic or age classification of the population, but the hospital will initially target the entire Jordanian market of persons who have health insurance, or at their own expense if they do not have health insurance. Continued local population growth will provide adequate demand for these services as well as demand from other countries. And thus taking care of those coming from outside Jordan in order to obtain appropriate treatment.

The project execution period represents the period following the decision to start the project. Large amounts of money will be spent in the planning and licensing process and until completion of construction and installation of the facilities.

Profit margin in the private health sector in Jordan is higher than in the public sector (approximately 35%), due to the high prices and the high value services provided by the private sector. In general, the difference between the profit margin of patients who are admitted is higher than those associated with outpatient review. Table 5 shows the basic profits of a private local hospital. The additional services available include pharmaceutical insurance, laboratory tests, radiation and other basic products necessary for effective medical treatment.

Table 4 Percentages of Basic Components of Private Hospital Profits

Pharmaceuticals	26.64%
Laboratories	13.44%
Medical and non - medical procedures	11.72%
Radiation Medicine	8.79%
Residence	7.6%
Operations rooms	6.93%
Surgery	5.24%
Emergency and doctors	4.91%
Internal medicine	3.47%
Respiratory diseases	2.95%
Delivery Service	2.08%
Dialysis Unit	1.9%
Dentistry	1.57%
Fertilization unit in vitro	1.14%
Pediatrics	0.88%
Physiotherapy	0.46%
nuclear medicine	0.12%
Lasik	0.07%
Research and training	0.07%
Total	100%

Jordan Enterprise Development Corporation

1.1 Competitors Analysis

After the study conducted by the team to identify competitors for the project, the existence of competitors, are Queen Rania Al-Abdullah government hospital and Ma'an government hospital where the total number of patients who received treatment in both hospitals is (253,274) patients. The following table represents the workload in both hospitals:

Table 5 the workload in Ma'an hospitals

Hospital name	No. of beds	Number of entries	Number of departures		Mortality rate	Number of sick days	Occupancy rate (Day)
			Alive	Dead			
Ma'an Male	-	2707	2640	67	2.5	15882	5.9
Female	-	5531	5473	58	1	16834	3
Total	131	8238	8113	125	1.5	32716	4
Male	-	1485	1439	47	3.2	3711	2.5
Queen Rania Female	-	2005	1974	30	1.5	4580	2.3
Total	72	3490	3413	77	2.2	8291	2.4

Ministry of Health 2015

Capacity of the project

The following tables represent the capacity of the hospital based on the availability of 115 beds for the first ten years:

Table 6 capacity of the hospital

	Capacity	Occupancy rate
Bed occupancy	36000	50%
Intensive care rooms occupancy	5400	50%

	Number of patients
Operations	10000
Emergency + outpatient clinics	30000
Laboratories	40000
Pharmacy	35000

As can be seen from the tables, based on market and competitors analysis, the total number of patients per year is expected to be approximately 42,000, knowing that the numbers in the tables represent the total use, regardless of the large proportion that the same patients who may have an operation or go to the emergency will use the laboratory, pharmacy and so on.

Price Analysis and Pricing Policy

During the first years of the project, the project will be based on locals, while the development and upgrading of services to international standards will be undertaken. The following is an explanation of the domestic prices that were relied upon in the feasibility analysis of the project:

- The rate of return for the surgical cases of patients admitted per bed per night was assumed to be about JD 1,000 per case. These fees cover room costs, medical procedures and treatment, as well as laboratory tests.
- It was assumed that the average rate of return per stay per night in the standard room is 50 JD, which represents the weighted average of stay.
- It was assumed that the rate of return per stay per night in the ICU room is 160 Jordanian Dinars, which represents the weighted average of stay.
- The average rate of return for each patient visit was assumed to be 30 JD, which represents the weighted average of the visit from the usual clinical fees in private hospitals (20 JD), in addition to treatments and medicines (10 JD).
- It was assumed that the rate of return for laboratories is 50 JD, which represents the weighted average for using laboratories.
- It was assumed that the rate of return of the pharmacy is 10 JD per patient.

It should be noted that these rates do not represent selling prices, but are average revenues derived from fees in private hospitals and linked with different types of accommodation and services. The following are prices for some of the main services of the Cardiac Surgery Hospital as an example of prices for some of the surgeries:

Table 7 sample of surgeries prices

Operation	Price (JD)
-----------	------------

Operation	Price (JD)
Regular residence / family	50
Accommodation / intensive care beds	160
Cardiac surgery	10,000
Catheterization of the heart	3,000
Cardiac catheterization	700
Radiography	10
Radiography (Fluorescope)	50
Tomography	80
Ultrasound wave scan	50
MRI	130
ECG	10
ECG ultrasound	50
Stress test	50
Heartbeat Recording	60

Prices were assumed by using price data for a local hospital and by calibrating and measuring revenue through published financial statements of a local hospital. Services prices were assumed which include services provided by the hospital such as medicines, laboratories, radiology, medical and non-medical procedures, Beds costs, etc. These revenues were used to analyze expected returns.

In the long term, hospital prices will be lower than the prices charged by private hospitals in Amman for similar services. The hospital will be able to increase its prices by providing excellent medical services and relying on local labor.

Table 8 Hospital capacity

	Capacity	Occupancy rate
Bed occupancy	36000	50%
Intensive care rooms occupancy	5400	50%

Projected Revenues

The following tables show the projected revenues for the project:

Table 9 projected revenues during the years (2018 - 2027)

	Number of patients	Expenditure rate	Total
Operations	10000	1000	10000000
Emergency + outpatient clinics	30000	30	900000
Laboratories	40000	50	2000000
Pharmacy	35000	10	350000
			13250000

	Capacity	Occupancy rate	Price	Revenues
Bed occupancy	36000	50%	50	900000
Intensive care rooms occupancy	5400	50%	160	432000
Total				1332000
Cafeteria revenues				24000

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Operating	1458200	1487364	1517111	1547453	1578402	1609970	1642170	1675013	1708513	1742684
Other revenues	24000	24480	24970	25469	25978	26498	27028	27568	28120	28682
Total revenue	1460600	1489812	1519608	1550000	1581000	1612620	1644872	1677770	1711325	1745552

3. Technical analysis

Required Human Resources

Human resources will be the most important input to the specialized services provided by the hospital to patients, as well as to specialized doctors and surgeons who are considered essential to the success of the proposed hospital.

The hospital needs 168 employees. Given the low number of doctors needed by the hospital and the relatively large number of experienced specialists and others in the field of health care in Jordan, human resources will not be an obstacle to the success of this project.

Jordan is distinguished from other countries because of its doctors, Jordan has about 17,319 doctors and other specialists working in the medical field. The following table shows the number of human cadres required for the project:

Table 10 the required labor for the project

Title	No.	Monthly Salary (JD)
General Manager	1	4,000
Director of Administrative Affairs	1	2,500
Director of Medical Management	1	2,500
Director of Nursing Department	1	2,500
Financial Manager	1	2,500
Director of Information Management	1	2,500
Director of Marketing and Public Relations	1	2,500
Radiologists	3	1,000
Anesthetists	3	1,000
Doctors – Revive Unit	3	1,000
Doctors - Physiotherapy Unit	6	1,500
Emergency Doctors	3	1,000
Doctors – Operations Unit	6	1,000
Sterilization Specialist	3	600
Director of Laboratories	1	1,500
Laboratory staff	10	350
Director of Pharmacy	1	1,200
Pharmacy staff	5	350

Title	No.	Monthly Salary (JD)
Medical Records Officer	3	450
Director of Nursing Affairs	1	1,200
Nurses	50	380
Accountants	5	400
Warehouses and Purchases	4	400
Information management staff	4	400
Marketing and public relations staff	6	400
Human resources staff	5	450
Maintenance staff and Public safety	5	350
Kitchen	6	350
Physiotherapy	10	300
Reception	6	300
Call center	2	300
Security	10	300
Total	168	

- **General Manager:** proposes the general policy of the hospital, preparing plans and following up on their implementation, technical and administrative supervision of employees, and developing procedures and securing the requirements of occupational safety and health.
- **Financial Manager:** Implementation of administrative, accounting and financial plans, systems and procedures, control of the Department's resources and expenses, organization of budget and accounts activities in the Department, preparation of financial reports, management of employees and development of their skills.
- **Manager of administrative affairs:** responsible for the administrative matters of the hospital and follows up on the work progress.
- **Manager of information management:** supervision of the database of the hospital and the re-development of medical systems, software and applications within the requirements.
- **Human resources manager:** Planning, staffing and working to measure their performance and develop their abilities and efficiency through effective training programs and work to motivate

these cadres to innovate and be creative in their areas of work and to organize the accumulation of experience and knowledge in the fields of work.

- **Laboratory Manager:** Planning, organizing, directing and monitoring all activities and services of the laboratory and blood bank.
- **Radiologist:** Responsible for radiology devices as well as films of radiation, chemicals and supplies, and prepare radiation reports.
- **General Surgery Specialist:** Provide all necessary medical services in the field of general surgery, in addition to providing necessary care to patients before and after the operation and supervision of all the necessary treatment and preventive services.
- **Anesthetist:** providing medical services in the field of anesthesia and recovery within the approved standards of the profession and the examination of oxygen and medical gases.
- **Doctors of ICU/CCU unit:** providing services to patients at the best level of quality and follow-up on cases according to the approved standards.
- **Laboratory technician:** taking samples and performing the required tests requested by the doctors and making sure that the tests are carried out according to the quality standards.
- **Nursing Officer:** Provide comprehensive nursing care and nursing plans that must follow and implement an effectively prepared nursing plan.
- **Nurse:** Provide care for outpatients and in hospital rooms.
- **Accountant:** Follow-up on financial matters and auditing.
- **Guard:** Maintain order and security in the hospital.
- **Maintenance Technician:** Perform the electrical and mechanical maintenance procedures and activities, and follow up on the implementation of the planned plans and programs so as to ensure the continuity of work of devices and their quality with maximum capacity and lowest possible cost, and follow up on the provision of spare parts and provide technical reports.
- **Public Safety Officer:** Examine the potential risks in the hospital and work to provide a safe work environment.
- **Warehouses and purchases:** Organize all operations related to medicines, consumables and devices, taking into consideration the proper functioning of these operations according to the regulations.
- **Cleaner:** maintaining the cleanliness of the hospital, offices and patient rooms.

Site Analysis

The proposed project will be established in Ma'an because of the lack of sufficient number of hospitals to serve the population, in addition to the provision of specializations that are not available in other hospitals in Ma'an with the modern medical equipment necessary for surgical operations and treatment of patients and others. In addition to the proximity of the border countries to facilitate the arrival of patients for treatment. It will be built on an area of 6000 square meters, and here are some of the Ministry of Health's requirements for site selection:

- The hospital should be in an area far from noise and pollution.
- The hospital should be close to residential areas for rapid response.
- Preferably in an area with good ventilation and away from industrial areas.
- Preferably close to clean water sources to ensure continuous water availability.
- It should be away from locations of high speed winds or near windbreaks.

Description of Services

The hospital will provide comprehensive services related to the diagnosis and treatment of various diseases, both for inpatient and outpatient clinics, with a focus on advanced clinical specialties that will distinguish the services provided by the hospital from those provided in other hospitals. The services will include diagnosing and treating cardiovascular diseases, as well as interventional radiology, catheterization, cardiac surgery, physical rehabilitation and general surgery, emergency treatments for various cases and many other services. As for clinical specialties, it includes many fields including:

- Department of Cardiology.
- Department of eye surgery.
- Department of Orthopedics.
- Department of Urology.
- Plastic Surgery Department.
- Pediatric Surgery Department.
- General Surgery Department.
- Physiotherapy Department.
- Emergency Department.
- ICU/CCU Department.

The hospital will meet the human needs of treatments for bone and joint diseases as well. There are three categories of individuals who need orthopedic treatments: those with congenital malformations in the bones and joints, and those who develop such cases from some of the occasional factors over time, and finally those who need emergency treatments as a result of accidents and injuries. The second category represents the largest of the three categories.

The hospital will provide health care services in the field of orthopedic surgery, arthritis and rehabilitation for each of the previous categories. These services include computer assisted surgeries and microscopic surgery to replace hip, knee, foot and ankle, as well as sports medicine, shock and tear treatments, paediatrics, physical medicine and other orthopedic treatments. The hospital will also address cases of referrals related to general, gynecological, obstetrics and joint surgery requiring specialized medical care. All this will be done by specialists and trained doctors to use the latest procedures such as pre-surgery consultations and specialized treatments.

Technical Requirements

The hospital needs to be registered with the required government authorities in addition to obtaining the necessary official licenses and approvals for set-up.

Outpatient visits and treatments for patients with admission and surgery are the source of annual hospital revenue. Additional services supporting the hospital include pharmaceutical, emergency, laboratory, endoscopy, x-ray and other for effective medical treatment.

1.1.1 Land and Construction

The project will be built on a land of 6000 square meters. The following table shows the required land:

Table11 Required land for the project

Land	Area sq.m
The required land	6000

The following table shows the required construction work for the project:

Table 12 Required Construction Work

Construction work	Area (m ²)
Building works	12,000
Building design and blueprints	12,000

Construction work	Area (m ²)
Digging	2,000
Paving	1,000
Soil examination	1

Machinery and Equipment

The following table shows the required machinery and equipment needed for the project:

Table 13 Machinery and Equipment

Machinery and equipment	Quantity
Patient rooms with their appliances	100
Operating rooms with their appliances	7
Laboratories	3
Department of ophthalmology	1
Department of Urology	1
Orthopedic Surgery Department	1
Plastic Surgery Department	1
Pediatric Surgery Department	1
General Surgery Department	1
ICU / CCU Room Appliances	15
Patient monitoring systems	10
Department of physiotherapy and cardiac with their appliances	1
Endoscopic surgical telescope	4
Emergency rooms with their appliances	1
IT Department	1
Reception	1
Meeting rooms	2
Ambulances	4
Equipment for washing	1
Kitchen	1
Dining room	1

Furniture

Table 14 the required furniture for the project

Furniture	Quantity
administrative Offices	100
Administrative Chairs	200
Central air conditioner	1
Washers	5
Central fire fighting device	1

Vehicles and Transportation

The following table shows a list of vehicles and means of transportation required for the project

Table 15 Vehicles and Transportation

Vehicles and Transportation	Quantity	Unit cost (JD)
Ambulances	4	100,000

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.

- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 16 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 17 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 18 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 19 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	45.00%
Utilities Expenses from Total Revenues	5.00%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.50%
Marketing Expenses from Total Revenues	1.00%

Table 20 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	20%
Income Tax Deduction(2)	80%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Risk Premium	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premium (3)	10.12%
Income Tax Rate	5.00%
Beta (4)	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 21 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	9,186,891
Loans Value	13,780,337
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	13,780,337
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	9,186,891
Equity Percentage	%40
Gross	
Project Value	22,967,229
Proportion of the project	%100
Weighted Average Cost of Capital	%12.05

Capital Expenditures

The estimated cost of the project is 22,967,229 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 22 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost
Required Land	50	6,000	300,000
Cost			300,000

Table 23 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost Jordan /
Building works	250	12,000	3,000,000
Building design and blueprints	7	12,000	84,000
Digging	5	2,000	10,000
Paving	12	1,000	12,000
Soil examination	200	1	200
Other works	-	-	310,620
Total			3,416,820

Table 24 Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantit	Cost / JD	Total Cost
Patient rooms with their appliances	100	5,000	500,000
Operating rooms with their appliances	7	100,000	700,000
Laboratories	3	150,000	450,000
Department of ophthalmology	1	300,000	300,000
Department of Urology	1	250,000	250,000
Orthopedic Surgery Department	1	300,000	300,000
Plastic Surgery Department	1	200,000	200,000
Pediatric Surgery Department	1	420,000	420,000
General Surgery Department	1	400,000	400,000
ICU / CCU Room Appliances	15	25,000	375,000
Patient monitoring systems	10	50,000	500,000
Department of physiotherapy and cardiac with their appliances	1	150,000	150,000
Endoscopic surgical telescope	4	35,000	140,000

Machinery and equipment	Quantit	Cost / JD	Total Cost
Emergency rooms with their appliances	1	200,000	200,000
IT Department	1	500,000	500,000
Reception	1	50,000	50,000
Meeting rooms	2	100,000	200,000
Ambulances	4	100,000	400,000
Equipment for washing	1	110,000	110,000
Kitchen	1	150,000	150,000
Dining room	1	25,000	25,000
Total (JD)			6,320,000

Table 25 Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
administrative Offices	100	200	20,000
Administrative Chairs	200	75	15,000
Central air conditioner	1	100,000	100,000
Washers	5	5,000	25,000
Central fire fighting device	1	75,000	75,000
Others	-	-	23,500
Total			258,500

Table 26 Transporation and Vehicles Capex

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pick up	4	100,000	400,000
Total			400,000

Table 27 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	11,500
Transportation	800
Marketing	1,500
Training	2,500
Total	16,300

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 28 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	12	%100	8,354,632
General and administrative expenses	12	%100	76,587
Marketing expenses	12	%100	146,060
Indirect Salaries	12	%100	1,590,400
Total			10,167,679

Table 29 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	300,000	10%	330,000	%1.4
Construction works	3,416,820	10%	3,758,502	%16.4
Machinery & Equipment	6,320,000	10%	6,952,000	%30.3
Furniture	258,500	10%	284,350	%1.2
Vehicles	400,000	10%	440,000	%1.9
Pre-Operating Expenses	16,300	10%	17,930	%0.1
Working capital	10,167,679	10%	11,184,447	%48.7
Total (JD)	20,879,299		22,967,229	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (22,967,229) JD, where 60% of the project will be financed through loans (13,780,337) JD, and the remaining (40%) through (Equity) with (9,186,891) JD. The following table summarizes the general structure of the required financing:

Table 30 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	9,186,891	%40
Loans	13,780,337	%60
Total	22,967,229	%100

Use of Fund	Value	Ratio
Capital Expenditures	11,764,852	%51.22
Pre-Operating Expenses	17,930	%0.08
Working Capital	11,184,447	%48.70
Total	22,967,229	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 31 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials	6,572,700	6,704,154	6,838,237	6,975,002	7,114,502	7,256,792	7,401,928	7,549,966	7,700,966	7,854,985
Utilities	730,300	744,906	759,804	775,000	790,500	806,310	822,436	838,885	855,663	872,776
Maintenance	146,060	148,981	151,961	155,000	158,100	161,262	164,487	167,777	171,133	174,555
Disposables	146,060	148,981	151,961	155,000	158,100	161,262	164,487	167,777	171,133	174,555
Others	759,512	774,702	790,196	806,000	822,120	838,563	855,334	872,441	889,889	907,687
Total	8,354,632	8,521,725	8,692,159	8,866,002	9,043,322	9,224,189	9,408,673	9,596,846	9,788,783	9,984,559

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 32 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	3,600	3,708	3,819	3,934	4,052	4,173	4,299	4,428	4,560	4,697
Hospitality	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Stationary and Printing	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Transportation	3,600	3,708	3,819	3,934	4,052	4,173	4,299	4,428	4,560	4,697

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Insurance	58,824	60,589	62,407	64,279	66,207	68,193	70,239	72,346	74,517	76,752
Miscellaneous	6,962	7,171	7,386	7,608	7,836	8,071	8,314	8,563	8,820	9,084
Total	76,587	78,884	81,251	83,688	86,199	88,785	91,449	94,192	97,018	99,928

Marketing Expenses

Table 33 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	7,817	9,058	10,064	11,071	12,077	13,084	14,090	15,097	16,103	17,110

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 34 Expected staff preparation

Job title	No.	Monthly	Total monthly salaries	Total yearly
General Manager	1	4,000	4,000	64,000
Director of Administrative Affairs	1	2,500	2,500	40,000
Director of Medical Management	1	2,500	2,500	40,000
Director of Nursing Department	1	2,500	2,500	40,000

Job title	No.	Monthly	Total monthly salaries	Total yearly
Financial Manager	1	2,500	2,500	40,000
Director of Information Management	1	2,500	2,500	40,000
Director of Marketing and Public Relations	1	2,500	2,500	40,000
Human resources staff	5	450	2,250	36,000
Maintenance staff and Public safety	5	350	1,750	28,000
Security	10	300	3,000	48,000
Kitchen	6	350	2,100	33,600
Physiotherapy	10	300	3,000	48,000
Reception	6	300	1,800	28,800
Call center	2	300	600	9,600
Radiologists	3	1,000	3,000	48,000
Anesthetists	3	1,000	3,000	48,000
Doctors – Revive Unit	3	1,000	3,000	48,000
Doctors - Physiotherapy Unit	6	1,500	9,000	144,000
Emergency Doctors	3	1,000	3,000	48,000
Doctors – Operations Unit	6	1,000	6,000	96,000
Sterilization Specialist	3	600	1,800	28,800
Director of Laboratories	1	1,500	1,500	24,000
Laboratory staff	10	350	3,500	56,000

Job title	No.	Monthly	Total monthly salaries	Total yearly
Director of Pharmacy	1	1,200	1,200	19,200
Pharmacy staff	5	350	1,750	28,000
Medical Records Officer	3	450	1,350	21,600
Director of Nursing Affairs	1	1,200	1,200	19,200
Nurses	50	380	19,000	304,000
Accountants	5	400	2,000	32,000
Warehouses and Purchases	4	400	1,600	25,600
Information management staff	4	400	1,600	25,600
Marketing and public relations staff	6	400	2,400	38,400
Grand Total	168		99,400	1,590,400

Table 35 Expected monthly salaries (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
General Manager	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Director of Administrative Affairs	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Director of Medical Management	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Director of Nursing Department	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Financial Manager	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Director of Information Management	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Director of Marketing and Public Relations	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Human resources staff	450	468	488	508	528	550	573	596	621	646
Maintenance staff and Public safety	350	364	379	395	411	428	445	464	483	502
Security	300	312	325	338	352	367	382	397	414	431
Kitchen	350	364	379	395	411	428	445	464	483	502
Physiotherapy	300	312	325	338	352	367	382	397	414	431
Reception	300	312	325	338	352	367	382	397	414	431
Call center	300	312	325	338	352	367	382	397	414	431
Radiologists	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Anesthetists	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Doctors – Revive Unit	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Doctors - Physiotherapy Unit	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Emergency Doctors	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Doctors – Operations Unit	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Sterilization Specialist	600	625	650	677	705	734	764	795	827	861
Director of Laboratories	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Laboratory staff	350	364	379	395	411	428	445	464	483	502
Director of Pharmacy	1,200	1,249	1,300	1,354	1,409	1,467	1,527	1,590	1,655	1,723
Pharmacy staff	350	364	379	395	411	428	445	464	483	502
Medical Records Officer	450	468	488	508	528	550	573	596	621	646

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Director of Nursing Affairs	1,200	1,249	1,300	1,354	1,409	1,467	1,527	1,590	1,655	1,723
Nurses	380	396	412	429	446	465	484	503	524	546
Accountants	400	416	433	451	470	489	509	530	552	574
Warehouses and Purchases	400	416	433	451	470	489	509	530	552	574
Information management staff	400	416	433	451	470	489	509	530	552	574
Marketing and public relations staff	400	416	433	451	470	489	509	530	552	574

Table 31 Annual Expected Salaries

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
General Manager	64,000	66,624	69,356	72,199	75,159	78,241	81,449	84,788	88,264	91,883
Director of Administrative Affairs	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Director of Medical Management	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Director of Nursing Department	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Financial Manager	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Director of Information Management	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Director of Marketing and Public Relations	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Human resources staff	36,000	37,476	39,013	40,612	42,277	44,010	45,815	47,693	49,649	51,684
Maintenance staff and Public safety	28,000	29,148	30,343	31,587	32,882	34,230	35,634	37,095	38,616	40,199
Security	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Kitchen	33,600	34,978	36,412	37,905	39,459	41,076	42,761	44,514	46,339	48,239
Physiotherapy	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912
Reception	28,800	29,981	31,210	32,490	33,822	35,208	36,652	38,155	39,719	41,347
Call center	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Radiologists	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912
Anesthetists	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912
Doctors – Revive Unit	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912
Doctors - Physiotherapy Unit	144,000	149,904	156,050	162,448	169,108	176,042	183,260	190,773	198,595	206,737
Emergency Doctors	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912
Doctors – Operations Unit	96,000	99,936	104,033	108,299	112,739	117,361	122,173	127,182	132,397	137,825
Sterilization Specialist	28,800	29,981	31,210	32,490	33,822	35,208	36,652	38,155	39,719	41,347
Director of Laboratories	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Laboratory staff	56,000	58,296	60,686	63,174	65,764	68,461	71,268	74,190	77,231	80,398
Director of Pharmacy	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Pharmacy staff	28,000	29,148	30,343	31,587	32,882	34,230	35,634	37,095	38,616	40,199
Medical Records Officer	21,600	22,486	23,408	24,367	25,366	26,406	27,489	28,616	29,789	31,011
Director of Nursing Affairs	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Nurses	304,000	316,464	329,439	342,946	357,007	371,644	386,881	402,744	419,256	436,446
Accountants	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Warehouses and Purchases	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Information management staff	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Marketing and public relations staff	38,400	39,974	41,613	43,319	45,096	46,945	48,869	50,873	52,959	55,130
Grand Total	1,590,40	1,655,60	1,723,48	1,794,14	1,867,70	1,944,28	2,024,00	2,106,98	2,193,37	2,283,30
	0	6	6	9	9	5	1	5	2	0

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 36 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation	Annual Additions
Land	330,000	%0.0	%0.0
Construction Works	3,758,502	%5.0	%0.0
Machinaries	6,952,000	%10.0	%2.0
Furniture	284,350	%10.0	%5.0
Vehicles	440,000	%10.0	%0.0
	11,764,852		

Table 37 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	11,764,852	11,918,110	12,074,859	12,235,191	12,399,200	12,566,982	12,738,639	12,914,274	13,093,992	13,277,906
Total Depreciation Expenses	955,560	970,886	986,561	1,002,594	1,018,995	1,035,773	1,052,939	1,070,502	1,088,474	1,106,865
Total Accumulated Depreciation	955,560	1,926,446	2,913,007	3,915,601	4,934,596	5,970,369	7,023,308	8,093,810	9,182,284	10,289,149
Total Additions	0	153,258	156,749	160,332	164,009	167,783	171,657	175,634	179,719	183,913
Total Net Book Values	10,809,292	9,991,664	9,161,852	8,319,590	7,464,604	6,596,614	5,715,332	4,820,464	3,911,709	2,988,756

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 38 Loan Details

Loan Value	13,780,337
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	3,587,054
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				13,780,337
2019	3,587,054	1,306,376	2,280,678	11,499,659
2020	3,587,054	1,090,168	2,496,886	9,002,773
2021	3,587,054	853,463	2,733,591	6,269,182
2022	3,587,054	594,318	2,992,735	3,276,447
2023	3,587,054	310,607	3,276,447	0

Income Statement

Table 39 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	14,606,000	14,898,120	15,196,082	15,500,004	15,810,004	16,126,204	16,448,728	16,777,703	17,113,257	17,455,522
Gross Operating Revenues	14,606,000	14,898,120	15,196,082	15,500,004	15,810,004	16,126,204	16,448,728	16,777,703	17,113,257	17,455,522
Operating Expenses	-8,354,632	-8,521,725	-8,692,159	-8,866,002	-9,043,322	-9,224,189	-9,408,673	-9,596,846	-9,788,783	-9,984,559
Gross Operating Profit	6,251,368	6,376,395	6,503,923	6,634,002	6,766,682	6,902,015	7,040,056	7,180,857	7,324,474	7,470,963
<i>Gross Profit Percentage</i>	<i>%43</i>	<i>%43</i>	<i>%43</i>	<i>%43</i>	<i>%43</i>	<i>%43</i>	<i>%43</i>	<i>%43</i>	<i>%43</i>	<i>%43</i>
Salaries and Benefits (Indirect	-1,590,400	-1,655,606	-1,723,486	-1,794,149	-1,867,709	-1,944,285	-2,024,001	-2,106,985	-2,193,372	-2,283,300

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Staff)										
General and Administraive Expenses	-76,587	-78,884	-81,251	-83,688	-86,199	-88,785	-91,449	-94,192	-97,018	-99,928
Markeing Expenses	-146,060	-148,981	-151,961	-155,000	-158,100	-161,262	-164,487	-167,777	-171,133	-174,555
Pre Operating Expenses	-17,930									
Gross Indirect Expenses	-1,830,977	-1,883,472	-1,956,698	-2,032,838	-2,112,008	-2,194,332	-2,279,937	-2,368,954	-2,461,522	-2,557,783
Income before Interest, Depreciation and Tax	4,420,391	4,492,923	4,547,225	4,601,164	4,654,673	4,707,683	4,760,119	4,811,903	4,862,952	4,913,180
Fixed Assets Depreciations	-955,560	-970,886	-986,561	-1,002,594	-1,018,995	-1,035,773	-1,052,939	-1,070,502	-1,088,474	-1,106,865
Income before Tax and Interests	3,464,831	3,522,038	3,560,665	3,598,570	3,635,679	3,671,910	3,707,180	3,741,400	3,774,478	3,806,315
Bank Interests	-1,306,376	-1,090,168	-853,463	-594,318	-310,607	-0	-0	-0	-0	-0
Income Before Tax	2,158,455	2,431,870	2,707,202	3,004,252	3,325,071	3,671,910	3,707,180	3,741,400	3,774,478	3,806,315
Income Tax	-107,923	-121,593	-135,360	-150,213	-166,254	-183,595	-185,359	-187,070	-188,724	-190,316
Net Profit	2,050,532	2,310,276	2,571,842	2,854,039	3,158,818	3,488,314	3,521,821	3,554,330	3,585,754	3,615,999
Net Profit Percentage	%14	%16	%17	%18	%20	%22	%21	%21	%21	%21
Compulsory Reserves	-205,053	-231,028	-257,184	-285,404	-315,882	-348,831	-352,182	-355,433	-358,575	-361,600
Retained Earnings	1,845,479	3,924,728	6,239,385	8,808,021	11,650,957	14,790,440	17,960,079	21,158,976	24,386,155	27,640,554

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 40 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	2,050,532	2,310,276	2,571,842	2,854,039	3,158,818	3,488,314	3,521,821	3,554,330	3,585,754	3,615,999
Non-monetary adjustments										
Bank benefits	1,306,376	1,090,168	853,463	594,318	310,607	0	0	0	0	0
Consumption	955,560	970,886	986,561	1,002,594	1,018,995	1,035,773	1,052,939	1,070,502	1,088,474	1,106,865
Operating cash flows before changes in working capital	4,312,469	4,371,330	4,411,865	4,450,952	4,488,420	4,524,088	4,574,760	4,624,833	4,674,228	4,722,864
Shortage (increase) in inventory	-547,725	-10,955	-11,174	-11,397	-11,625	-11,858	-12,095	-12,337	-12,583	-12,835
Decrease (increase) in receivables	-1,217,167	-24,343	-24,830	-25,327	-25,833	-26,350	-26,877	-27,415	-27,963	-28,522
(Decrease) increase in accounts payable	296,989	5,940	6,059	6,180	6,303	6,429	6,558	6,689	6,823	6,959
Change in working capital	-1,467,903	-29,358	-29,945	-30,544	-31,155	-31,778	-32,414	-33,062	-33,723	-34,398
Net cash flows (used) from operating activities	2,844,566	4,341,972	4,381,920	4,420,407	4,457,265	4,492,309	4,542,346	4,591,771	4,640,505	4,688,467
Cash flows from investing activities										
(Purchase) of fixed assets	-11,764,852	-153,258	-156,749	-160,332	-164,009	-167,783	-171,657	-175,634	-179,719	-183,913
Net flows (used) from investment	-11,764,852	-153,258	-156,749	-160,332	-164,009	-167,783	-171,657	-175,634	-179,719	-183,913

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
activities										
Cash flows from financing activities										
Equity capital	9,186,891									
Loan Amortization	-2,280,678	-2,496,886	-2,733,591	-2,992,735	-3,276,447	0	0	0	0	0
Bank Interest Rate	-1,306,376	-1,090,168	-853,463	-594,318	-310,607	-0	-0	-0	-0	-0
Loans	13,780,337	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	19,380,175	-3,587,054	-3,587,054	-3,587,054	-3,587,054	0	0	0	0	0
activities										
Net increase (decrease) in cash	10,459,888	601,661	638,117	673,022	706,202	4,324,527	4,370,689	4,416,136	4,460,786	4,504,554
Cashflows at the Beginning of Period	0	10,459,888	11,061,549	11,699,666	12,372,687	13,078,889	17,403,416	21,774,105	26,190,242	30,651,028
Cashflows at the End of Period	10,459,888	11,061,549	11,699,666	12,372,687	13,078,889	17,403,416	21,774,105	26,190,242	30,651,028	35,155,582

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 41 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	10,459,888	11,061,549	11,699,666	12,372,687	13,078,889	17,403,416	21,774,105	26,190,242	30,651,028	35,155,582
Inventory	547,725	558,680	569,853	581,250	592,875	604,733	616,827	629,164	641,747	654,582
Accounts receivable	1,217,167	1,241,510	1,266,340	1,291,667	1,317,500	1,343,850	1,370,727	1,398,142	1,426,105	1,454,627
Total assets traded	12,224,780	12,861,738	13,535,859	14,245,605	14,989,265	19,351,999	23,761,660	28,217,547	32,718,880	37,264,790
Non-current assets										
Fixed assets (net)	10,809,292	9,991,664	9,161,852	8,319,590	7,464,604	6,596,614	5,715,332	4,820,464	3,911,709	2,988,756
Total non-current assets	10,809,292	9,991,664	9,161,852	8,319,590	7,464,604	6,596,614	5,715,332	4,820,464	3,911,709	2,988,756
Total assets	23,034,072	22,853,402	22,697,711	22,565,195	22,453,869	25,948,613	29,476,992	33,038,011	36,630,588	40,253,547
Liabilities	-	-	-	-	-					
Current liabilities										
Payables	296,989	302,928	308,987	315,167	321,470	327,899	334,457	341,147	347,970	354,929
Remaining amount of Loan	2,496,886	2,733,591	2,992,735	3,276,447	0-	0-	0-	0-	0-	0
Total current liabilities	2,793,875	3,036,519	3,301,722	3,591,613	321,470	327,899	334,457	341,147	347,970	354,929
Non-current liabilities										
long term loans	9,002,773	6,269,182	3,276,447	0	0	0	0	0	0	0
Total long - term liabilities	9,002,773	6,269,182	3,276,447	0	0	0	0	0	0	0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total liabilities	11,796,648	9,305,702	6,578,169	3,591,613	321,470	327,899	334,457	341,147	347,970	354,929
Property rights										
Member Contributions	9,186,891	9,186,891	9,186,891	9,186,891	9,186,891	9,186,891	9,186,891	9,186,891	9,186,891	9,186,891
Compulsory reserve	205,053	436,081	693,265	978,669	1,294,551	1,643,382	1,995,564	2,350,997	2,709,573	3,071,173
Profit (loss)	1,845,479	3,924,728	6,239,385	8,808,021	11,650,957	14,790,440	17,960,079	21,158,976	24,386,155	27,640,554
Total equity	11,237,424	13,547,700	16,119,542	18,973,581	22,132,399	25,620,713	29,142,534	32,696,865	36,282,619	39,898,618
Total liabilities and equity	23,034,072	22,853,402	22,697,711	22,565,195	22,453,869	25,948,613	29,476,992	33,038,011	36,630,588	40,253,547

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 42 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	-22,967,229	2,862,496	4,188,714	4,225,171	4,260,075	4,293,256	4,324,527	4,370,689	4,416,136	4,460,786	4,504,554	58,183,910
Discount Factor	1.00	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.29	0.26	0.26
Net Present Value for	-22,967,229	2,034,654	2,657,104	2,391,960	2,152,330	1,935,799	1,740,180	1,569,594	1,415,343	1,275,889	1,149,834	14,852,042

Table below (43) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 43 payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	22,967,229	2,862,496	4,188,714	4,225,17	4,260,07	4,293,25	4,324,527	4,370,689	4,416,136	4,460,78	4,504,554
Project Value Returned	22,967,229	20,104,73	15,916,01	11,690,8	7,430,77	3,137,51	1,187,010	5,557,699	-9,973,836	14,434,6	-18,939,176
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 44 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	10,207,500
Payback Period	5
Internal Rate of Return	%21.33

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 10,207,500 Jordan Dinars considering that the project provides 168 Job Opportunities for the governorate residents.

Table 45 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%21.33	5	12.05%
Revenues declined by 10%	%19.10	6	12.05%
Operating Expenses Increased by 10%	%17.04	7	12.05%